

OULU VIHREASAARI, Finland

WMO: 028760

Lat: 65.006N

Lon: 25.393E

Elev: 12

StdP: 14.69

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.0	-8.1	-18.9	2.0	-13.8	-12.4	2.8	-8.0	34.8	34.1	29.7	29.7	5.0	140	0.759

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.7	76.1	65.8	72.7	64.6	69.8	62.8	68.2	74.1	65.8	71.0	63.8	68.9	9.7	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.9	95.5	71.5	63.5	87.7	69.3	61.4	81.5	67.2	32.5	74.3	30.6	71.0	29.1	69.0	74.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
29.5	25.4	22.3	-15.4	80.1	9.1	5.4	-21.9	83.9	-27.2	87.1	-32.3	90.1	-38.8	94.0
			-15.6	69.7	8.5	3.8	-21.7	72.4	-26.6	74.7	-31.4	76.8	-37.5	79.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	38.6	15.5	18.1	23.5	33.6	47.0	55.8	62.7	59.2	51.3	39.4	31.3	23.9
	DBStd	17.92	12.24	13.00	9.79	6.02	8.13	6.36	5.84	5.12	5.36	5.96	7.46	10.74
	HDD50	5196	1069	893	821	491	156	16	0	2	49	331	560	808
	HDD65	9713	1534	1313	1286	941	560	285	114	191	411	795	1010	1273
	CDD50	1025	0	0	0	0	63	190	395	288	88	1	0	0
	CDD65	68	0	0	0	0	3	9	44	12	0	0	0	0
	CDH74	227	0	0	0	0	16	27	152	32	0	0	0	0
	CDH80	40	0	0	0	0	0	5	33	2	0	0	0	0
Wind	WSAvg	11.1	10.5	11.1	11.2	9.9	11.3	11.9	9.9	10.5	11.4	12.2	12.0	11.8
Precipitation	PrecAvg	23.5	1.9	1.6	1.4	1.3	1.7	2.1	2.9	2.4	2.1	2.2	2.2	2.0
	PrecMax	31.6	3.2	3.2	2.2	3.0	3.2	4.0	5.2	5.6	3.8	3.6	3.8	3.7
	PrecMin	17.8	0.7	0.2	0.5	0.4	0.4	0.3	0.4	0.3	0.4	0.6	0.5	0.6
	PrecStd	4.4	0.7	0.8	0.5	0.7	0.7	0.9	1.3	1.3	1.0	0.9	0.9	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.2	40.1	42.2	53.5	76.7	77.8	84.2	78.7	67.0	53.4	46.9	38.7
		MCWB	35.1	37.0	37.4	46.4	58.7	65.3	67.2	68.2	59.7	50.0	44.3	36.9
	2%	DB	34.5	37.2	38.7	46.8	68.0	71.7	77.6	72.6	62.9	51.0	44.2	37.4
		MCWB	33.3	35.1	35.2	41.3	56.4	61.7	67.1	65.5	57.3	48.5	42.7	35.6
	5%	DB	33.1	35.0	36.9	43.0	62.6	68.0	74.1	69.2	60.3	49.0	41.8	36.2
		MCWB	32.3	33.5	34.1	38.6	54.1	60.5	65.9	62.9	56.3	46.7	40.5	34.7
	10%	DB	31.5	33.3	35.3	40.8	57.7	64.6	71.4	66.8	58.6	47.3	39.6	34.8
		MCWB	30.8	32.2	33.3	37.6	51.1	58.3	64.4	61.3	55.1	45.1	38.5	33.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.5	37.2	37.9	48.4	60.2	66.3	71.6	69.8	60.9	51.9	44.8	37.5
		MCDB	36.8	39.2	41.1	52.6	71.0	72.5	76.7	75.4	65.7	52.5	46.3	38.4
	2%	WB	33.7	35.5	35.9	42.0	57.3	64.1	69.4	66.8	58.5	48.8	42.8	36.0
		MCDB	34.2	37.3	38.0	46.3	65.7	70.0	75.2	72.2	61.7	50.3	44.2	37.0
	5%	WB	32.5	33.8	34.7	39.6	54.5	61.6	67.3	64.0	57.0	47.2	40.9	34.9
		MCDB	33.2	34.8	36.4	42.1	62.0	67.0	72.8	68.2	59.7	48.9	41.6	35.9
	10%	WB	30.6	32.6	33.5	38.0	51.8	59.1	65.3	62.0	55.6	45.4	38.7	33.9
		MCDB	31.4	33.5	35.1	40.4	56.7	64.0	70.6	65.6	58.3	47.2	39.4	34.7
Mean Daily Temperature Range	5% DB	MDBR	10.3	10.8	12.1	9.8	10.4	9.1	9.7	9.4	8.2	6.7	6.0	7.8
		MCDBR	8.8	7.8	9.1	12.2	18.0	14.5	14.2	12.8	10.9	7.3	6.9	6.2
		MCWBR	8.6	7.1	7.3	7.9	9.4	8.1	8.5	7.7	6.9	6.0	6.1	5.5
	5% WB	MCDBR	8.8	6.7	8.3	10.9	16.3	12.7	12.5	10.7	9.4	7.3	6.3	6.0
		MCWBR	8.6	6.2	6.9	7.9	9.5	7.9	8.3	7.3	6.4	6.4	5.9	5.4
Clear-Sky Solar Irradiance	taub	0.209	0.278	0.309	0.346	0.348	0.351	0.370	0.354	0.331	0.307	0.344	0.355	
	taud	1.952	2.108	2.161	2.241	2.425	2.468	2.436	2.473	2.519	2.356	2.551	2.606	
	Ebn at Noon	93	189	240	259	273	274	264	258	239	184	86	36	
	Edh at Noon	7	19	28	34	31	30	30	27	21	15	7	4	
All-Sky Solar Radiation	RadAvg	34	196	664	1201	1574	1764	1628	1205	660	240	50	10	
	RadStd	3	38	67	96	149	140	147	113	67	39	8	1	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (39 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page